

WASP-12b

R-1818

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_160201 · created 2026-08-03T04:26:40+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2457418.70268 +/- 0.00024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.142 +/- 0.024
Transit depth (Rp/Rs)^2	2.02 +/- 0.7 [%]
Orbital Inclination (inc)	86.75 +/- 2.46
Airmass coefficient 1 (a1)	1.121 +/- 0.021
Airmass coefficient 2 (a2)	-0.102 +/- 0.049
Scatter in the residuals of the lightcurve fit is	1.85 %
Best Comparison Star	#1 - [491, 393]
Optimal Aperture	3.55
Optimal Annulus	7.11
Transit Duration (day)	0.1315 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.142 ± 0.024 vs 0.1178 (deviation 1.01σ)
Duration fit vs published	0.1315 d vs 0.125 d (deviation 1.63σ)
Mid-time O–C	-1571.5 min
Depth SNR	5.9
Residual scatter	1.85 %

Light curve

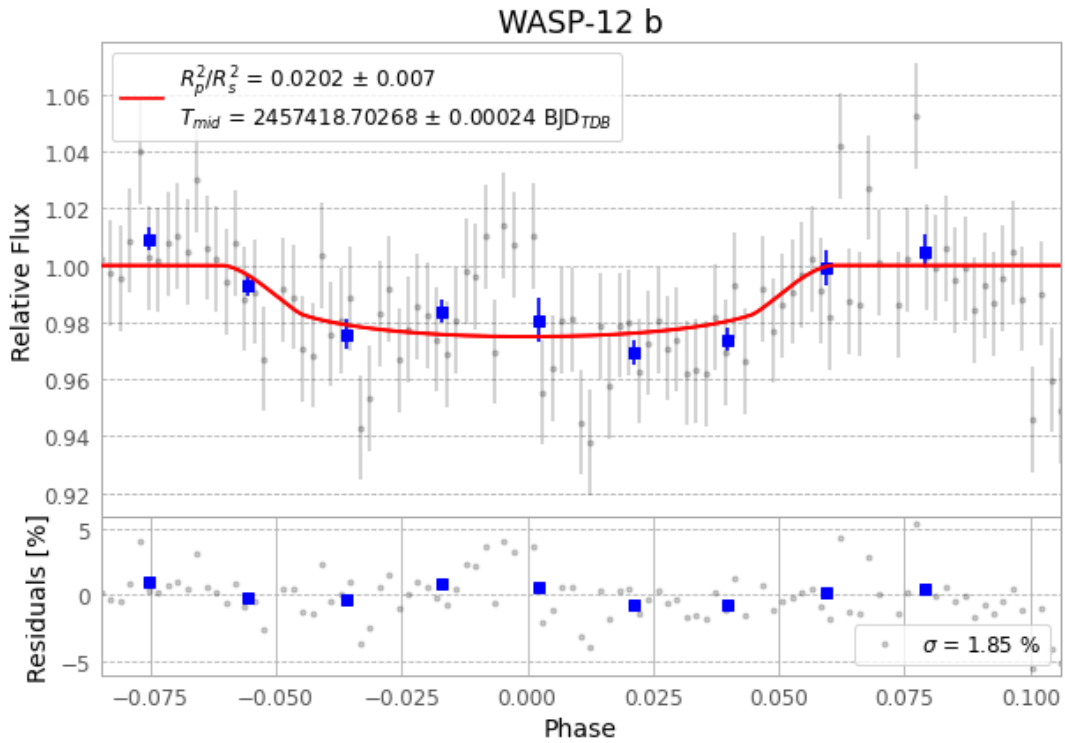


Figure 1 — FinalLightCurve_WASP-12 b_2016-01-30.png (SHA-256 8ef2f8b52128e5bd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 213], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 070661a5be655660...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

101 FITS frames (66 MB), WASP-12160131023012.FITS → WASP-12160131073012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160201.log (2c1963fc2694...), FinalLightCurve_WASP-12 b_2016-01-30.png (8ef2f8b52128...), FinalLightCurve_WASP-12 b_2016-01-30.csv (0de098b07558...)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 04:26Z.